

Foundation of Mathematics

1- Summary for Numbers, Equations, Inequalities and Basic functions

I. Numbers

❖ Set Of Numbers

- **Natural numbers:** $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- **Whole numbers:** $\mathbb{N}^* = \{0, 1, 2, 3, 4, \dots\}$
- **Integers:** $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- **Rational numbers:**
 $\mathbb{Q} = \left\{\frac{n}{m}; n, m \in \mathbb{Z}; m \neq 0\right\}$
- **Irrational numbers:**
 $\bar{\mathbb{Q}} = \{x: x \notin \mathbb{Q}\} = \{\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi, e, \dots\}$
- **Real numbers:** $\mathbb{R} = \mathbb{Q} \cup \bar{\mathbb{Q}}$

$$\mathbb{N} \subseteq \mathbb{N}^* \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

When can a decimal number be written in a rational form?

1. If it is an Integer number.
2. If the number of digits after the decimal point is **FINITE; 2.35**.
3. If the number of digits after the decimal point is **INFINITE** but, with a **REPEATED PERIOD; 2.35**.

❖ Converting Decimal Number into a Rational Form:

- i. Separate the un-repeated part from the repeated part and make the repeated part starts just after the decimal point.
- ii. Rename the repeated part as (p) and multiply (p) by 10 raised to a power "number of digits in the repeated digits".
- iii. Solve to get p, substitute in (i), and common denominator to get the final rational form.

Example: $0.11\overline{25} = 0.11 + 0.00\overline{25}$
 $= 0.11 + \left(\frac{1}{100} * 0.\overline{25}\right)$

Let $p = 0.\overline{25} \Rightarrow 100p = 25.252525\dots$
 $\Rightarrow = 25 + 0.252525\dots$
 $\Rightarrow = 25 + p$
 $\Rightarrow p = \frac{25}{99}$

So, $0.11\overline{25} = 0.11 + \left(\frac{1}{100} * 0.\overline{25}\right)$
 $= 0.11 + \left(\frac{1}{100} * \frac{25}{99}\right)$
 $= \frac{11}{100} + \frac{25}{9900} = \frac{557}{4950}$

II. Equations

Eq. Type	General Form	Solution
Linear	$ax + b = c$	$x = \frac{c - b}{a}$
Quadratic	$ax^2 + bx + c = 0$	✓ Factorization ✓ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ✓ Complete Square Method
Containing Absolute	$ x = a$	$\begin{cases} x = \pm a & \text{if } a > 0 \\ x = 0 & \text{if } a = 0 \\ \text{No solution} & \text{if } a < 0 \end{cases}$ Note: ✚ $u = v \Rightarrow u = \pm v$ ✚ $\sqrt{x^2} = x \neq x$ ✚ $\sqrt{u(x)^2} = u(x)$

Note:

Complete Square Method for Quadratic Equations

- i. $ax^2 \pm bx + c \dots$ (take "a" common factor)
- ii. $a\left(x^2 \pm \frac{b}{a}x + \frac{c}{a}\right)$
- iii. $a\left[\left(x \pm \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a}\right]$

III. Inequalities

- 1) If $x < y$ then,
 - a) $Kx < Ky$... if $K > 0$
 - b) $Kx > Ky$... if $K < 0$
- 2) $|u| < k \Rightarrow -k < u < k \Rightarrow \begin{cases} u < k \\ \text{AND} \\ u > -k \end{cases}$
- 3) $|u| > k \Rightarrow \begin{cases} u > k \\ \text{OR} \\ u < -k \end{cases}$
- 4) $u^2 < k \Rightarrow |u| < \sqrt{k} \Rightarrow \text{back to (2)}$
- 5) $u^2 > k \Rightarrow |u| > \sqrt{k} \Rightarrow \text{back to (3)}$
- 6) $|u| < |k| \Rightarrow u^2 < k^2$
- 7) $|u| > |k| \Rightarrow u^2 > k^2$ } in case we have 2 absolutes
- 8) $|u| > u \Rightarrow u < 0$
- 9) $\sqrt{u(x)^2} = |u(x)|$
- 10) $\sqrt{x^2} = |x| = \begin{cases} x & ; x \geq 0 \\ -x & ; x < 0 \end{cases}$
- 11) $a < u^2 < b \Rightarrow \sqrt{a} < |u| < \sqrt{b}$
 $\Rightarrow \sqrt{a} < u < \sqrt{b} \text{ OR } -\sqrt{a} > u > -\sqrt{b}$
- 12) $a < c < b \Rightarrow (\text{Squaring})$
 $\Rightarrow 0 < c^2 < \max\{a^2; b^2\}$
- 13) $b > a$ where a and b are reals $\Rightarrow \frac{1}{b} < \frac{1}{a}$
 if a, b are (+ve) together or (-ve) together
- 14) (i) "A or B" means " $A \cup B$ ".
 (ii) "A and B" means " $A \cap B$ ".

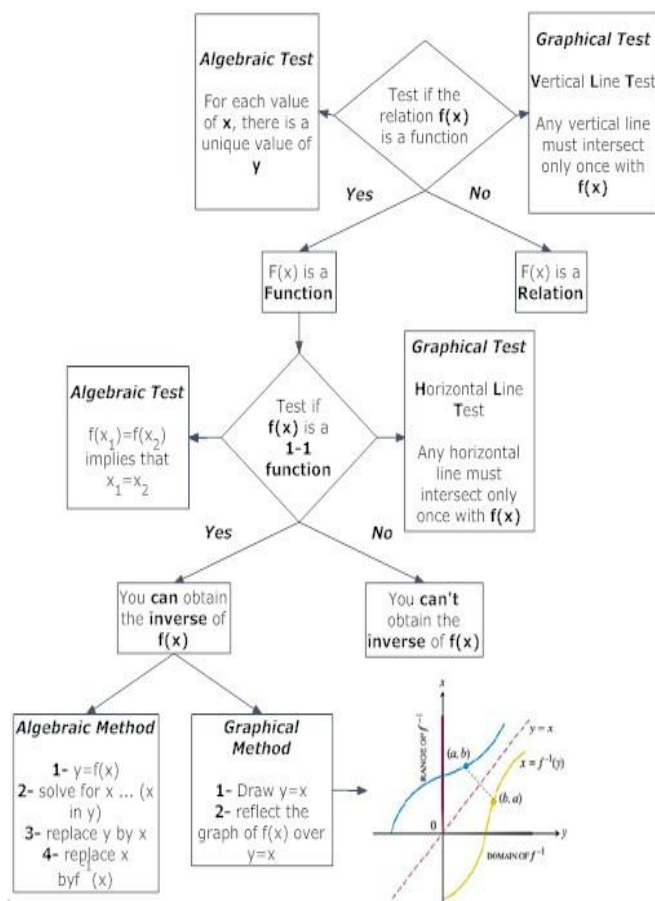
Note:

- Multiplying any inequality by (-ve) sign changes its sign ($>$ or $<$).
- Multiplying any inequality $\frac{u(x)}{f(x)} < \text{or} > v(x)$ by function $f(x)$, results in having 2 cases:

$$\begin{cases} u(x) < \text{or} > v(x) * f(x) & \text{if } f(x) > 0 \\ \text{OR} \\ u(x) > \text{or} < v(x) * f(x) & \text{if } f(x) < 0 \end{cases}$$

IV. Basic Functions

- ❖ Function VS Relation
- ❖ Check 1-1 Function
- ❖ Get the inverse of $f(x)$



❖ Sketching Rules and Steps

1. Reflection :

$$f(x) \Rightarrow \begin{cases} f(-x) & \dots \text{reflect on } \dots y\text{-axis} \\ -f(x) & \dots \text{reflect on } \dots x\text{-axis} \end{cases}$$

2. Vertical Shifting : $f(x) \Rightarrow f(x) + k$

$$\text{if ; } k = \begin{cases} +ve & \dots \text{Shift } \dots \text{Up} \\ -ve & \dots \text{Shift } \dots \text{Down} \end{cases}$$

3. Horizontal Shifting :

$$(i) f(x) \Rightarrow f(x \pm k), \text{ Put } x \pm k = 0:$$

$$\text{if ; } x = \begin{cases} +ve & \dots \text{Shift } \dots \text{Right} \\ -ve & \dots \text{Shift } \dots \text{Left} \end{cases}$$

$$(ii) f(-x) \Rightarrow f(-x \pm k), \text{ Put } -x \pm k = 0$$

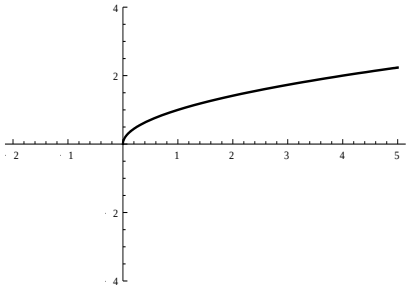
The order of sketching using the previous steps is:

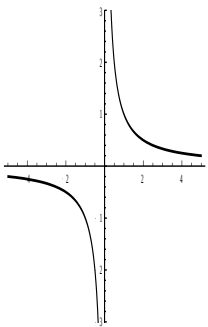
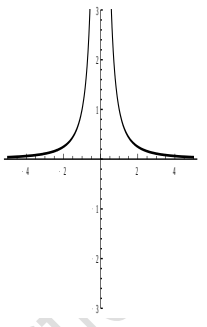
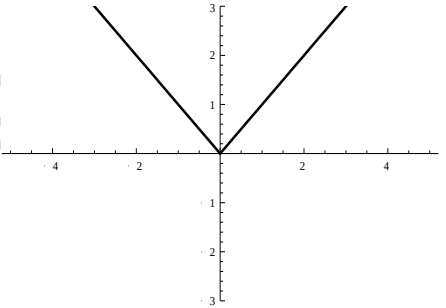
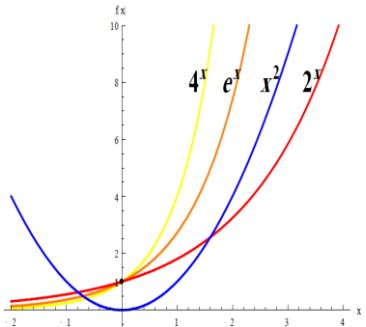
- i. Draw the basic function.
- ii. Reflection on y-axis (if required).
- iii. Horizontal shift.
- iv. Reflection on x-axis (if required).
- v. Vertical shift.

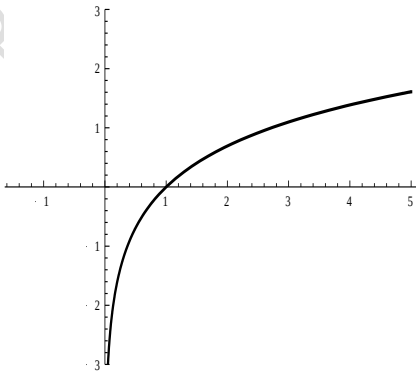
Note:

- **Domain** of a function $f(x)$ is defined as the set of all possible values of x that can be substituted in $f(x)$.
- **Range** of a function $f(x)$ is defined as the set of all values of $f(x)$ "y" obtained after substituting with the domain of x .
- To get the range of functions **algebraically**: start from the domain ($a < x < b$), apply mathematical operations to find the range ($c < f(x) < d$).
- To say that $f(x) = g(x)$, they must have the same (Result when sub. With any value of x) and the same (Domain). **OR** if a function $f(x)$ can be simplified to equal another $g(x)$, but they don't have the same domain $\Rightarrow f(x) \neq g(x)$
- **Cancellation rule: (between any function 'f' and its inverse f^{-1})**
 - $f(f^{-1}(x)) = x : x \in D_{f^{-1}} = R_f$
 - $f^{-1}(f(x)) = x : x \in R_{f^{-1}} = D_f$
- $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$... **But**, they have different domains.

❖ Types of Functions

Function Type	Domain (D)	Well Known Example	Domain OR Graph + Notes
1- Polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0$	$\mathbb{R}$	$f(x) = x, x^2, x^3$	
2- Radical (Square root) (its output ≥ 0) $f(x) = \sqrt{g(x)}$ Range: $y \geq 0$ or $[0, \infty)$	Domain of square root function obtained by: $g(x) \geq 0$	$G(x) = \sqrt{x}$ & $f(x) = \sqrt{x^2 - 4}$	 $D_f: x^2 - 4 \geq 0$ $x^2 \geq 4 \Rightarrow x \geq 2$ $D_f = \mathbb{R} -] - 2, 2[$

3- Rational $\frac{g(x)}{h(x)}$, where g, h are polynomials	$\mathbb{R} - \{h(x) = 0\}$	$f(x) = \frac{x^2 - 9}{x - 1}$	1/x  1/x²  $D_f = \mathbb{R} - \{1\}$
4- Absolute value “Modulus” (its output ≥ 0) $f(x) = g(x) $ Range: $y \geq 0$ or $[0, \infty)$ x is symmetric about $x = 0$ "y-axis" $ax + b$ is symmetric about $x = -\frac{b}{a}$	Domain of absolute function obtained by: Domain of $g(x)$	$f(x) = x $	
5- Exponential (its output > 0) $f(x) = a^{g(x)}$,where a is a positive real number (except $a=1$) If: $a=1 \dots f(x) = 1^x = 1$,where x is any real number Range: $y > 0$ or $(0, \infty)$	Domain of exponential function equals: Domain of $g(x)$	$f(x) = e^x$	$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828$ $e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$ e^x is called the natural exponential function Properties of e^x $e^a * e^b = e^{a+b}$ $\frac{e^a}{e^b} = e^{a-b}$ $e^{-a} = \frac{1}{e^a}$ $(e^a)^b = e^{ab} = e^{ba}$ $e^{\ln x} = x ; x > 0$ 

6- Logarithmic $f(x) = \log_a g(x)$ where a is positive number except $a=1$ ($\log_a x$ is the inverse function of a^x) $y = \log_a x \Leftrightarrow a^y = x$	Domain of logarithmic function obtained by: $g(x) > 0$	$f(x) = \ln x$ where $\ln = \log_e$	$\log_e(x) = \ln(x)$ is the natural logarithmic function. $\log_{10}(x) = \log(x)$ is the common logarithmic function. Properties of $\ln x$ 1) $\ln(a * b) = \ln a + \ln b$ 2) $\ln \frac{a}{b} = \ln a - \ln b$ 3) $\ln \frac{1}{a} = -\ln a$ 4) $\ln a^b = b * \ln a$ 5) $\ln e^x = x ; x \in \mathbb{R}$ 6) $\log_a x = \frac{\ln x}{\ln a}$  In general: $a^{\log_a x} = x \dots \text{if } x > 0$ $\log_a a^x = x \dots \text{if } x \in \mathbb{R}$
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❖ Equation of straight line

- The standard form of the linear function $f(x) = ax + b$
- The point slope from $y - y_1 = m(x - x_1) \dots$ point $p(x_1, y_1)$ is on the line with slope m
- The slope intercept form $y = mx + b \dots$ Line with slope m and y-intercept $(0, b)$
- The slope of a non-vertical straight line is defined by $m = \frac{y_2 - y_1}{x_2 - x_1}$
- The **vertical lines** have **no slopes** but, the **horizontal lines** have **zero slopes**.
- If two lines are **parallel**, then $m_1 = m_2$
- If two lines are **perpendicular**, then $m_1 \cdot m_2 = -1$

❖ Summary of Domains

- $D(f \pm g) = D(f) \cap D(g)$ **and** $D(f * g) = D(f) \cap D(g)$.
 - If f is a polynomial, then $D_f = \mathbb{R}$
 - if $f(x) = \frac{1}{\sqrt{g(x)}}$, then D_f is obtained from putting: $g(x) > 0$.
 - if $f(x) = \frac{p(x)}{q(x)}$ is a rational function, then $D_f = (D_p \cap D_q) - \{x: q(x) = 0\}$.
 - if $f(x) = \frac{1}{q(x)}$, then $D_f = \mathbb{R} - \{x: q(x) = 0\}$.
 - if $f(x) = \sqrt{u(x)}$, then D_f is obtained from putting: $u(x) \geq 0$
 - if $f(x) = \frac{1}{\sqrt{u(x)}}$, then D_f is obtained from putting: $u(x) > 0$
 - if $f(x) = e^{u(x)}$, then $D_f =$ the domain of $u(x)$
 - if $f(x) = \log_a u(x)$, then D_f is obtained from putting: $u(x) > 0$
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7- Trigonometric Functions

(Sin, Cos, Tan, Cot, Sec, Csc)

$$\sin(\theta) = \frac{y}{r}$$

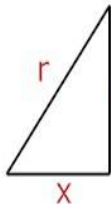
$$\cos(\theta) = \frac{x}{r}$$

$$\tan(\theta) = \frac{y}{x}$$

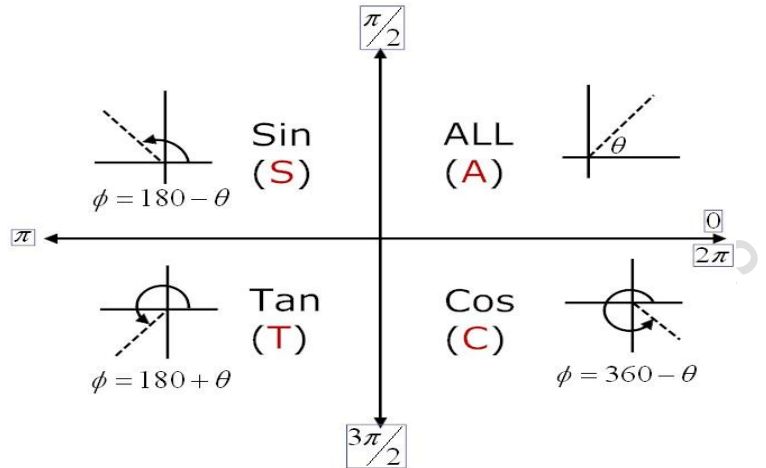
$$\cot(\theta) = \frac{x}{y}$$

$$\sec(\theta) = \frac{r}{x}$$

$$\csc(\theta) = \frac{r}{y}$$

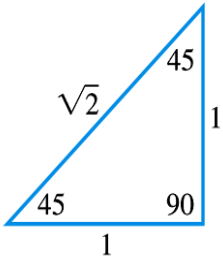
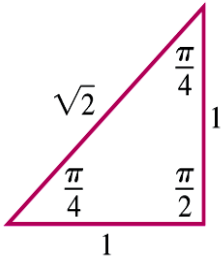
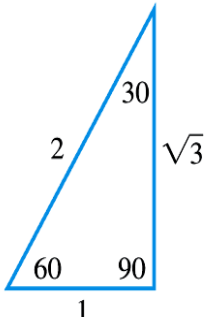
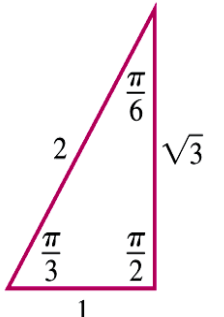


y: Opposite
x: Adjacent
r: Hypotenuse



Function	Inverse	Function	Inverse
Sin(x) 	Sin⁻¹(x) 	Cos(x) 	Cos⁻¹(x)
D = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ R = [-1, 1]	D = [-1, 1] R = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	D = [0, pi] R = [-1, 1]	D = [-1, 1] R = [0, pi]
Tan(x) 	Tan⁻¹(x) 	Cot(x) 	Cot⁻¹(x)
D = $\left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$ R = $]-\infty, \infty[$	D = $]-\infty, \infty[$ R = $\left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$	D = $]0, \pi[$ R = $]-\infty, \infty[$	D = $]-\infty, \infty[$ R = $]0, \pi[$
Sec(x) 	Sec⁻¹(x) 	Csc(x) 	Csc⁻¹(x)
D = $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ R = $]-\infty, \infty[-]-1, 1[$	D = $]-\infty, \infty[-]-1, 1[$ R = $[0, \pi] - \left\{\frac{\pi}{2}\right\}$	D = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$ R = $]-\infty, \infty[-]-1, 1[$	D = $]-\infty, \infty[-]-1, 1[$ R = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Common angles x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin x$	$\frac{1}{2}\sqrt{0}$	$\frac{1}{2}\sqrt{1}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{4}$
$\cos x$	$\frac{1}{2}\sqrt{4}$	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{1}$	$\frac{1}{2}\sqrt{0}$

Degrees	Radians
	
	

$$e \approx 2.718 \quad \dots \quad \pi \neq \frac{22}{7} \quad \dots \quad \pi \approx \frac{22}{7} \quad \dots \quad \pi \cong 3.14$$

❖ **Identities of the Trigonometric Functions:**

- $\sin^2(x) + \cos^2(x) = 1$
- $1 + \tan^2(x) = \sec^2(x)$
- $1 + \cot^2(x) = \csc^2(x)$
- $\cos(a \pm b) = \cos(a) * \cos(b) \mp \sin(a) * \sin(b)$
- $\sin(a \pm b) = \sin(a) * \cos(b) \pm \sin(b) * \cos(a)$
- $\sin(2x) = 2 * \sin(x) * \cos(x)$
- $\cos(2x) = \cos^2(x) - \sin^2(x) = 1 - 2\sin^2(x) = 2\cos^2(x) - 1 \Rightarrow \begin{cases} \cos^2(x) = \frac{1}{2}(1 + \cos(2x)) \\ \sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \end{cases}$
- $\tan(a \pm b) = \frac{\tan(a) \pm \tan(b)}{1 \mp \tan(a) * \tan(b)}$
- $\sin(-x) = -\sin(x) \dots \text{Odd function}$
- $\cos(-x) = \cos(x) \dots \text{Even function}$
- $\tan(-x) = -\tan(x) \dots \text{Odd function}$

❖ **Note that:** if given $f(x) = x * \sin(x)$, and we need to find for example $f\left(\frac{\pi}{2}\right)$?

Then, $f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} * \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} * \sin(90^\circ)$, this means that the external " x " which is not an angle must be written in radians " $\frac{\pi}{2}$ ", while the internal " x " which is an angle must be written in degrees " 90° ".

❖ **Periodicity of trigonometric functions**

Def: Periodic Function: A function $f(x)$ is periodic if there is a positive number p , such that $f(x+p)=f(x)$ for every value of x . the smallest such value of p is the period of f .

Note : For $\sin(ax)$ & $\cos(bx)$... The period is $\frac{2\pi}{a}$

Note : For $\sin(ax + c)$ & $\cos(bx + c)$... The period is $\frac{2\pi}{a}$

Note : For $|\sin(ax)|$ & $|\cos(bx)|$... The period is $\frac{\pi}{a}$

❖ **Identities of the Inverse Trigonometric Functions:**

1. $\sin(\sin^{-1}(x)) = x \quad \dots \quad -1 \leq x \leq 1$
2. $\sin^{-1}(\sin(x)) = x \quad \dots \quad \frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$
3. $\cos(\cos^{-1}(x)) = x \quad \dots \quad -1 \leq x \leq 1$
4. $\cos^{-1}(\cos(x)) = x \quad \dots \quad 0 \leq x \leq \pi$
5. $\tan(\tan^{-1}(x)) = x \quad \dots \quad -\infty < x < \infty$
6. $\tan^{-1}(\tan(x)) = x \quad \dots \quad \frac{-\pi}{2} < x < \frac{\pi}{2}$
7. $\csc^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$
8. $\sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$
9. $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right)$
10. $\sin^{-1}(-x) = -\sin^{-1}(x)$
11. $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$
12. $\tan^{-1}(-x) = -\tan^{-1}(x)$
13. $\cos^{-1}(x) + \sin^{-1}(x) = \frac{\pi}{2}$

$$-1 \leq \sin(u(x)) \leq 1$$

$$-1 \leq \cos(u(x)) \leq 1$$

$$\frac{-\pi}{2} < \tan^{-1}(u(x)) < \frac{\pi}{2}$$

$$0 \leq \sin^2(u(x)) \leq 1$$

$$0 \leq \cos^2(u(x)) \leq 1$$

❖ **Note: if** $-1 \leq x \leq 1$ then $\sin(\sin^{-1}(x)) = x$ **BUT** if $x \notin [-1, 1]$ then $\sin(\sin^{-1}(x)) = \text{UNDEFINED}$.

❖ **Note: if** $\frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$ then $\sin^{-1}(\sin(x)) = x$

❖ BUT if $x \notin \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ then, you have to find the reference angle of " x " first to be able to apply the cancellation rule.

❖ **To find the reference angles:** it is better to go like:

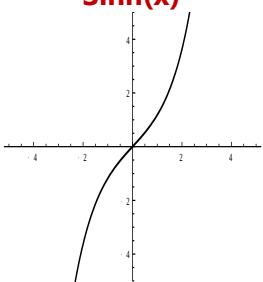
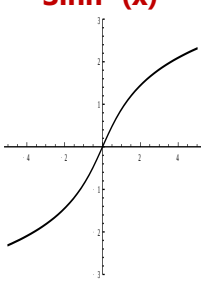
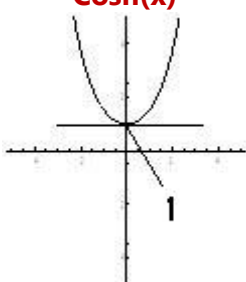
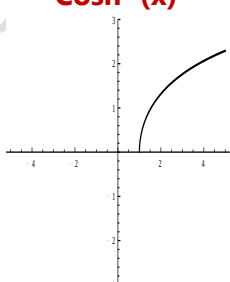
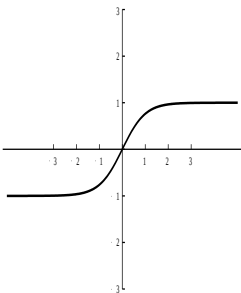
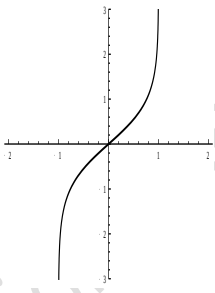
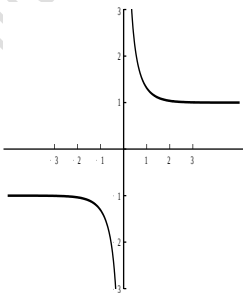
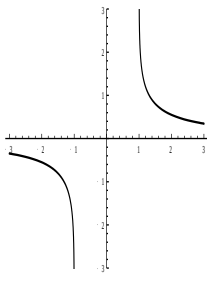
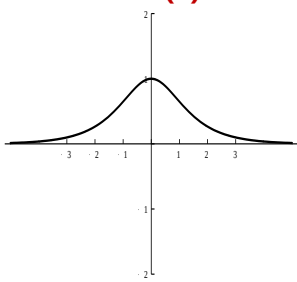
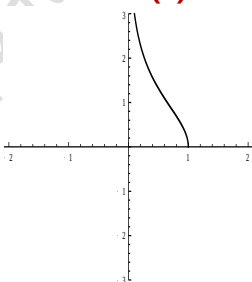
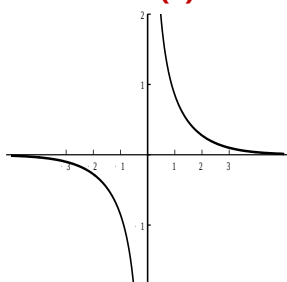
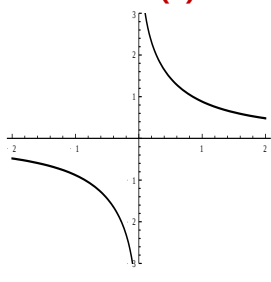
- If x is an angle in the 2nd quad, then $x = 180 - \theta$,
- If x is an angle in the 3rd quad, then $x = 180 + \theta$,
- If x is an angle in the 4th quad, then $x = 360 - \theta$.

Where, θ is an angle in the 1st quad and then use the ASTC rules, or the addition rules.

8- Hyperbolic Functions (Sinh, Cosh, Tanh, Coth, Sech, Csch)

Definition of Hyperbolic functions:-

$\sinh(x) = \frac{e^x - e^{-x}}{2}$	$\operatorname{csch}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^x - e^{-x}}$
$\cosh(x) = \frac{e^x + e^{-x}}{2}$	$\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}}$
$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$\operatorname{coth}(x) = \frac{1}{\tanh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Function	Inverse	Function	Inverse
Sinh(x) 	Sinh⁻¹(x) 	Cosh(x) 	Cosh⁻¹(x) 
Tanh(x) 	Tanh⁻¹(x) 	Coth(x) 	Coth⁻¹(x) 
Sech(x) 	Sech⁻¹(x) 	Csch(x) 	Csch⁻¹(x) 

Domains of Hyperbolic Functions:-

Function	Domain	Function	Domain
Sinh (x)	$]-\infty, \infty[$	Csch(x)	$x \neq 0$
Cosh(x)	$]-\infty, \infty[$	Sech(x)	$]-\infty, \infty[$
Tanh(x)	$]-\infty, \infty[$	Coth(x)	$x \neq 0$

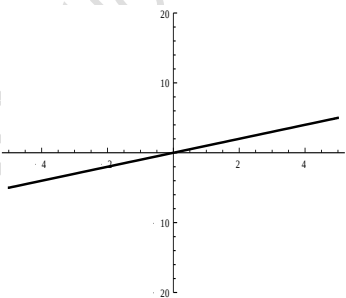
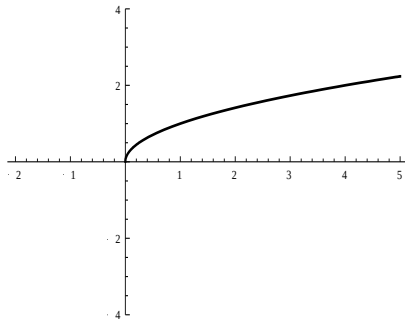
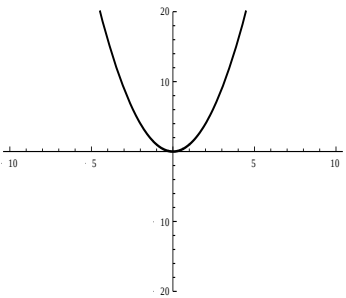
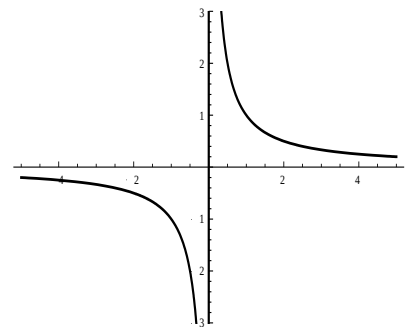
❖ **Identities of the Hyperbolic Functions:**

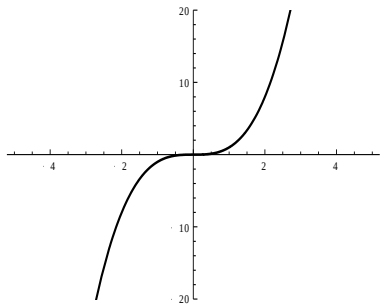
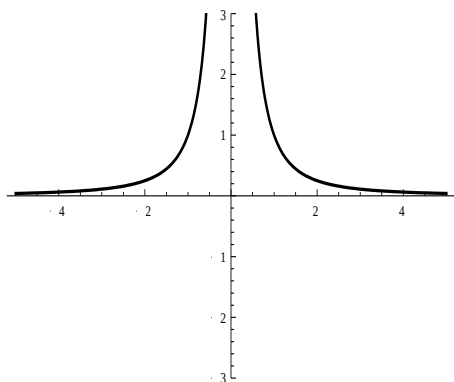
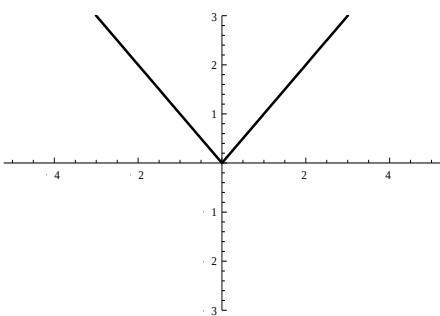
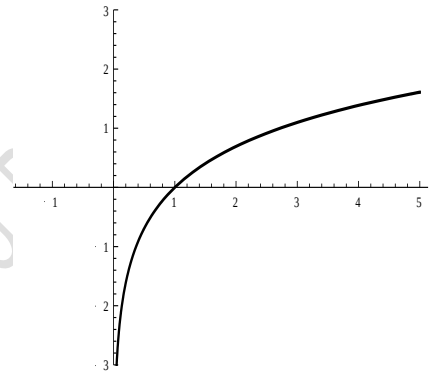
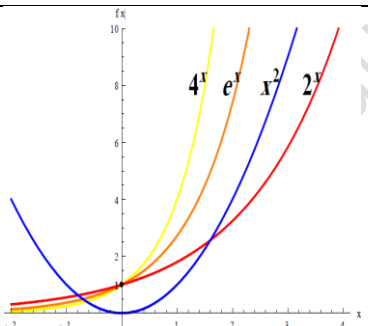
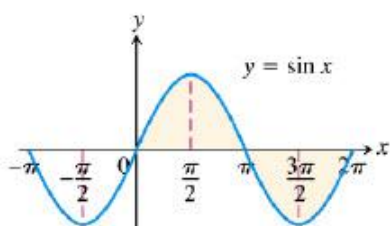
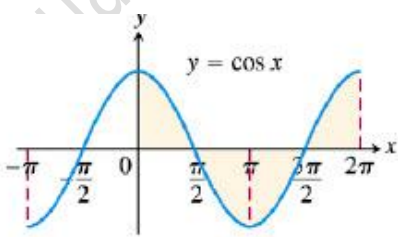
1. $\cosh^2(x) - \sinh^2(x) = 1$
2. $1 - \tanh^2(x) = \operatorname{sech}^2(x)$
3. $\coth^2(x) - 1 = \operatorname{csch}^2(x)$
4. $\sinh(2x) = 2 * \sinh(x) * \cosh(x)$
5. $\cosh(a \pm b) = \cosh(a) * \cosh(b) \pm \sinh(a) * \sinh(b)$
6. $\sinh(a \pm b) = \sinh(a) * \cosh(b) \pm \sinh(b) * \cosh(a)$
7. $\cosh(2x) = \cosh^2(x) + \sinh^2(x) = 1 + 2\sinh^2(x) = 2\cosh^2(x) - 1$
8. $-1 < \tanh(u(x)) < 1$
9. $\sinh(-x) = -\sinh(x)$... Odd function
10. $\cosh(-x) = \cosh(x)$... Even function

❖ **Identities of the Inverse Hyperbolic Functions:**

1. $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$
2. $\cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$... $x \geq 1$
3. $\tanh^{-1}(x) = \frac{1}{2} * \ln\left(\frac{1+x}{1-x}\right)$: $-1 < x < 1$
4. $\operatorname{sech}^{-1}(x) = \cosh^{-1}\left(\frac{1}{x}\right)$
5. $\operatorname{csch}^{-1}(x) = \sinh^{-1}\left(\frac{1}{x}\right)$
6. $\coth^{-1}(x) = \tanh^{-1}\left(\frac{1}{x}\right)$

❖ **Graphs of Basic functions to be memorized**

Function	Graph	Function	Graph
$y = x$		$y = \sqrt{x}$	
$y = x^2$		$y = \frac{1}{x}$	

$y = x^3$		$y = \frac{1}{x^2}$	
$y = x $		$y = \ln(x)$	
$y = e^x$		$y = \sin x$	
$y = \cos x$		$y = \tan x$	